Online Resource 1: Mathematical checks of reward-to-update claims

Yanqi Li
University of Southern Queensland
Yanqi.Li@unisq.edu.au

Supplement to *Enhancing Reinforcement Learning with Large Language Models: A Survey of Grounded Reward, Policy, and Environment Interfaces*, submitted to *Artificial Intelligence Review*.

This note gives the complete constructions behind the survey's two checks: preservation of discounted task preferences, and the passage from reward quality to a policy update. It connects each construction to a published formula in ARM-FM, SWE-RM, or ProCredit. The numerical results below are analytical examples constructed for the survey; they are not measurements from those papers' training runs. Two accompanying standard-library Python programs check the constructions: one reproduces the numerical examples by exact arithmetic and exhaustive enumeration; the other verifies the calibrated family using exact identities, fixed-seed numerical checks, finite differences, and independent group enumeration. Neither program trains a policy.

1 Discounted preferences under ARM-FM

Source condition and learning target. Appendix A.5 of Creus Castanyer et al. (2026) states optimal-policy preservation when a reward machine has no positive-reward cycles and its terminal reward exceeds the cumulative reward of every nonterminal trajectory. Its DQN implementation adds the task and machine rewards. With environment state s, machine state u, language embedding $\phi(u)$, next-state quantities s', u', terminal indicator d, and target network $\widehat{Q}$, Algorithm 1, line 20 (PDF p. 19), gives

$$y = r_{\text{task}} + r_{\text{RM}} + \gamma(1-d)\max_{a'} \widehat{Q}(s', \phi(u'), a'). \tag{1}$$

Construction. Consider a deterministic episodic task with initial state s, intermediate state x, and terminal goal g. One action reaches g immediately; another reaches x, from which the next action reaches g. The task pays one on entry to g and zero otherwise. Its discount is $\gamma = .99$.

The reward machine accurately recognizes entry to x and task completion. It accepts completion on either route; the intermediate event is optional for this task. Such optional events are permitted by the published formalism: the Meta-World machine includes both a direct $u_0 \to u_2$ grasp transition and a route through u_1 that earns an additional near-object reward (Creus Castanyer et al., 2026, Appendix A.9, PDF p. 32). The constructed machine pays $b = .2$ at x and one at g. There are no positive-reward cycles; unused events can have zero-reward self-loops. Every nonterminal reward total is at most .2, strictly below the terminal reward one. Both conditions in the source are satisfied.

Path	Task return	Machine return	Additive return
$s \to g$	1	1	2
$s \to x \to g$	.99	$.2 + .99 = 1.19$	$.2 + 2(.99) = 2.18$

The task prefers immediate completion, whereas the machine-only and additive objectives prefer the delayed path. At the Bellman fixed point of Eq. (1), the terminal target is two. The delayed

action therefore has value $.2 + .99(2) = 2.18$, above the direct action's value two. This is a comparison of the target's fixed-point preferences, not the first network update from an arbitrary initialization.

For a general intermediate bonus b, the machine-only reversal requires $b > 1 - \gamma$; the additive reversal requires

$$b > 2(1 - \gamma). \tag{2}$$

The source's requirement $b < 1$ does not exclude either. Both constructed paths succeed. The counterexample concerns discounted preference preservation, not a failed trajectory outranking completion, and it does not establish failure in the source's experimental tasks.

2 Calibration and a policy-gradient direction

2.1 The missing conditional covariance

For a trajectory τ, let $r(\tau)$ be a fixed predicted score, $c(\tau) \in \{0, 1\}$ its correctness, and $h(\tau) = \nabla_\theta \log \pi_\theta(\tau)$ its policy score function. Define $J_f(\theta) = \mathbb{E}_{\pi_\theta}[f]$. Appendix C.3 of Shum et al. (2026) states calibration without a uniform-in-policy qualification. At a fixed sampling policy, this condition is

$$\mathbb{E}[c \mid r] = r \tag{3}$$

From this condition, the source infers equality of $\mathbb{E}[rh]$ and $\mathbb{E}[ch]$ (source Eq. 12, PDF p. 16). Under the usual differentiability conditions for a fixed evaluator,

$$\begin{aligned} \nabla J_r - \nabla J_c &= \mathbb{E}[(r - c)h] \\ &= -\mathbb{E}[\mathrm{Cov}(c, h \mid r)]. \end{aligned} \tag{4}$$

To see the second equality, condition on r and use $\mathbb{E}[ch \mid r] = \mathbb{E}[c \mid r]\mathbb{E}[h \mid r] + \mathrm{Cov}(c, h \mid r)$. Calibration fixes the average correctness within a score group. It does not force the policy sensitivities of correct and incorrect trajectories in that group to agree.

2.2 An exactly calibrated three-action example

Consider a one-step task with three actions and independent softmax logits:

$$\pi_z(i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}, \qquad z_i^0 = \log p_i.$$

Use the following probabilities, correctness labels, and scores:

Action i	p_i	c_i	r_i
1	.1	1	1
2	.6	1	2/3
3	.3	0	2/3

The evaluator is exactly calibrated: the score-one action is correct, and correctness in the lower-score group is $.6/(.6 + .3) = 2/3$. Both expected score and success equal .7.

For fixed values f_i,

$$\partial_{z_i} J_f|_{z^0} = p_i(f_i - J_f). \tag{5}$$

Thus the success and score gradients are

$$g_c = (.03, .18, -.21), \qquad g_r = (.03, -.02, -.01). \tag{6}$$

Along an ordinary Euclidean logit step $z(\eta) = z^0 + \eta g_r$,

$$\left.\frac{dJ_r}{d\eta}\right|_0 = \|g_r\|^2 = .0014, \qquad \left.\frac{dJ_c}{d\eta}\right|_0 = g_c^\top g_r = -.0006. \tag{7}$$

A sufficiently small positive step increases score and decreases success. No later distribution shift is needed to obtain this local discrepancy. Calibration throughout a neighborhood of policies would instead imply equality of the two objectives there, and hence equality of their gradients. That is stronger than Eq. (3) at one policy.

2.3 A family of calibrated base policies

For $a, b, d > 0$ with $a + b + d = 1$, write $m = 1 - a = b + d$ and set

$$p = (a, b, d), \qquad c = (1, 1, 0), \qquad r = \left(1, \frac{b}{m}, \frac{b}{m}\right). \tag{8}$$

The score-one action is correct, and correctness conditional on the lower score is b/m, so the evaluator is exactly calibrated at p. For each chosen base point, freeze these scores and define $J_f(z) = \sum_i \pi_z(i) f_i$, where $\pi_z(i) = e^{z_i} / \sum_j e^{z_j}$ and $z_i^0 = \log p_i$. The gradients below are evaluated in the full three-logit space at z^0. They do not differentiate the rule that constructs r from the base point.

Since $J_c(z^0) = J_r(z^0) = 1 - d$ and $\partial_{z_i} J_f = p_i(f_i - J_f)$, direct calculation gives

$$\begin{aligned} g_c &= d\big(a, b, -(1-d)\big), \\ g_r &= ad\left(1, -\frac{b}{m}, -\frac{d}{m}\right), \\ g_c - g_r &= \frac{bd}{m}(0, 1, -1). \end{aligned} \tag{9}$$

Thus the gradient equality fails at every interior point in this family, including points where the two gradients still have a positive inner product. The stronger condition of a negative success derivative is

$$\begin{aligned} g_c^\top g_r &= \frac{ad^2}{m}\left[am - b^2 + d(1-d)\right] \\ &= \frac{ad^2}{1-a} B(a, d), \\ B(a, d) &= (1-a)(2a-1) + (3-2a)d - 2d^2. \end{aligned} \tag{10}$$

The prefactor is positive in the interior. Also $g_r \neq 0$, so an ordinary Euclidean step $z^0 + \eta g_r$ has $\dot{J}_r(0) = \|g_r\|^2 > 0$ and $\dot{J}_c(0) < 0$ precisely when $B(a, d) < 0$. Smoothness then gives the corresponding strict changes for all sufficiently small positive step sizes.

The complete strict negative region. For fixed a, B is a concave quadratic in d on $[0, 1-a]$. Its endpoint values are $(1-a)(2a-1)$ and $2a(1-a)$. For $a \geq 1/2$, both endpoints are nonnegative and the interior is positive. For $0 < a < 1/2$, the lower root lies inside this interval and the upper root lies beyond it. Hence the complete interior region is

$$0 < a < \frac{1}{2}, \qquad 0 < d < d_*(a) := \frac{3 - 2a - \sqrt{1 + 12a(1-a)}}{4}. \tag{11}$$

At $a = 1/10$,

$$d_*(1/10) = \frac{7 - \sqrt{13}}{10} = 0.3394448724536011\ldots. \tag{12}$$

For example, the open rectangle $a \in (.09, .11)$, $d \in (.29, .31)$ lies wholly in the negative region. Both partial derivatives of B are positive there, while its supremum at the upper corner is $B(.11, .31) = -.0246$. This establishes a set of positive two-dimensional measure in the base-policy construction, rather than an isolated numerical example. Across this two-dimensional set, the selected score $b/(1-a)$ can change. It is not a claim that one fixed evaluator remains calibrated on an open neighborhood of the full policy simplex, nor an estimate of how often a trained system encounters this region.

A fixed evaluator and a simpler parametrization. Put $q = b/(1-a)$, so that $p = (a, (1-a)q, (1-a)(1-q))$ and $r = (1, q, q)$. Equation (10) becomes

$$g_c^\top g_r = a(1-a)^2(1-q)^2\left[2a(q^2 - q + 1) - q(2q-1)\right]. \tag{13}$$

Consequently, its strict negative region can equivalently be written as

$$\frac{1}{2} < q < 1, \qquad 0 < a < \frac{q(2q-1)}{2(q^2 - q + 1)}. \tag{14}$$

In particular, fixing $q = 2/3$ fixes the evaluator throughout the family and reduces the condition to $0 < a < 1/7$:

$$g_c^\top g_r = \frac{2a(1-a)^2(7a-1)}{81}, \qquad \|g_r\|^2 = \frac{14a^2(1-a)^2}{81}. \tag{15}$$

The numerical example above corresponds to $a = .1$.

The distinction between the base-policy family and the update coordinates is essential. Along the curve with q fixed, $J_c(p(a)) = J_r(p(a)) = q + a(1-q)$, so both derivatives with respect to a are $1 - q$. Equivalently,

$$\frac{dz^0}{da} = \left(\frac{1}{a}, -\frac{1}{1-a}, -\frac{1}{1-a}\right), \qquad g_c^\top \frac{dz^0}{da} = g_r^\top \frac{dz^0}{da} = 1 - q.$$

The unfavorable direction in Eq. (15) is the ambient ordinary logit gradient, which need not be tangent to this calibrated curve. For example, it changes the low-score odds at rate $\log(p_2/p_3) = (g_r)_2 - (g_r)_3 = -ad(2q-1)$. For the negative region, $q > 1/2$, so those odds move against the correct low-score action even though the two low-score actions have identical predicted scores.

3 Connecting the family to SWE-RM's GSPO surrogate

3.1 The initial derivative

SWE-RM uses group sequence policy optimization (GSPO), a sequence-normalized ratio, and standardized group advantages (Shum et al., 2026, Appendix E.1, Eqs. 14–15, PDF pp. 20–21). For a group of G trajectories from the behavior policy, write

$$\rho_i(\theta) = \left[\frac{\pi_\theta(\tau_i)}{\pi_{\text{old}}(\tau_i)}\right]^{1/L_i}, \qquad A_i = \frac{r_i - \bar{r}}{s_G}, \tag{16}$$

where L_i is the positive trajectory-length factor in the source ratio, $\bar{r} = G^{-1}\sum_i r_i$, and s_G is the group standard deviation. The clipped surrogate averages $\min\{\rho_i A_i, \text{clip}(\rho_i, 1-\varepsilon, 1+\varepsilon)A_i\}$

over the group. At $\theta = \theta_{\text{old}}$, the ratio is one and clipping is inactive for a clipping interval of positive width. Holding the sampled advantages and behavior policy fixed gives

$$\nabla_\theta \widehat{J} = \frac{1}{G} \sum_{i=1}^{G} \frac{A_i}{L_i} \nabla_\theta \log \pi_\theta(\tau_i). \tag{17}$$

The source reports no KL loss and uses 16 rollouts per problem. The calculation below concerns this initial surrogate derivative with no additional loss gradient; it does not assume a particular optimizer implementation.

3.2 Equal-length groups over the calibrated family

Use the family in Section 2.3, with $q = b/(1-a)$, one generated token per trajectory, and $G \geq 2$ members sampled independently from the base policy. For this calculation let $A_i = (r_i - \bar{r})/(s_G + \epsilon_s)$ with $\epsilon_s > 0$. The source displays s_G without specifying a stabilizer or the treatment of constant-score groups. The positive stabilizer used here makes those groups contribute zero without an undefined division.

Let $N \sim \text{Binomial}(G, a)$ count score-one samples, $u = N/G$, and $\Delta = 1 - q > 0$. Take $t_G = 1$ for population standard deviation or $t_G = \sqrt{G/(G-1)}$ for sample standard deviation. The high- and low-score centered rewards are $\Delta(1-u)$ and $-\Delta u$, with common denominator $t_G \Delta \sqrt{u(1-u)} + \epsilon_s$. Since group advantages sum to zero, the common $-p$ terms in the softmax scores $e_i - p$ cancel. Conditional on N, the low-score actions occur in expected proportions q and $1-q$. Therefore

$$\begin{aligned}
\mathbb{E}[\widehat{g} \mid N] &= k(u)(1, -q, -(1-q)), \\
k(u) &= \frac{\Delta u(1-u)}{t_G \Delta \sqrt{u(1-u)} + \epsilon_s}, \\
\mathbb{E}[\widehat{g}] &= K_{G,\epsilon_s}(a, q)(1, -q, -(1-q)), \\
K_{G,\epsilon_s}(a, q) &= \sum_{n=0}^{G} \binom{G}{n} a^n (1-a)^{G-n} k(n/G) > 0.
\end{aligned} \tag{18}$$

Here $\widehat{g} = \nabla_z \widehat{J}$. Strict positivity follows because $0 < a < 1$ and $G \geq 2$ give positive probability to a mixed-score group. From Eq. (9),

$$\mathbb{E}[\widehat{g}] = \frac{K_{G,\epsilon_s}(a, q)}{ad} g_r. \tag{19}$$

Standardization thus changes the magnitude, but not the initial expected logit direction, throughout this two-score family. In particular, for the fixed evaluator $q = 2/3$,

$$g_c^\top \mathbb{E}[\widehat{g}] = \frac{2(1-a)(7a-1)}{27} K_{G,\epsilon_s}(a, 2/3) < 0 \quad \text{if } 0 < a < 1/7. \tag{20}$$

Removing the standard-deviation divisor leaves $A_i = r_i - \bar{r}$. The self-included group mean gives

$$\mathbb{E}[\widehat{g}_{\text{centered}}] = \frac{G-1}{G} g_r, \qquad \dot{J}_c = \frac{G-1}{G} g_c^\top g_r. \tag{21}$$

Indeed, write $\bar{r} = (r_i + \sum_{j \neq i} r_j)/G$. Independence makes the other members' contribution to the expected score-function product zero, leaving the coefficient $(G-1)/G$. The entire strict negative region in Eq. (14) therefore persists after removing standardization.

The original numerical example as a limit. For $a = .1$, $q = 2/3$, and population standard deviation, the limit $\epsilon_s \downarrow 0$ gives

$$\kappa_G := \lim_{\epsilon_s \downarrow 0} K_{G,\epsilon_s}(.1, 2/3) = \sum_{n=0}^{G} \binom{G}{n} (.1)^n (.9)^{G-n} \sqrt{\frac{n}{G}\left(1 - \frac{n}{G}\right)} > 0. \tag{22}$$

Constant-score groups contribute zero in this limit. Equivalently, these numbers follow from zero advantages on such groups and population standardization on mixed groups; no division by zero is performed. The limiting derivatives are

$$\dot{J}_c = -.02\kappa_G < 0, \qquad \dot{J}_r = \frac{7}{150}\kappa_G > 0. \tag{23}$$

G	κ_G	Standardized-limit $\dot{J}_c$	Centered-only $\dot{J}_c$
2	.0900000000	−.0018000000	−.0003000
8	.2094920147	−.0041898403	−.0005250
16	.2563031126	−.0051260623	−.0005625

Equal positive trajectory lengths add a common inverse-length factor to the update. Unequal lengths, nonindependent sampling, later clipped updates, or an additional loss gradient need not preserve the collinearity in Eq. (19).

3.3 Optimizer and parameterization boundaries

For a positive diagonal preconditioner $D = \mathrm{diag}(w_1, w_2, w_3)$, Eq. (10) becomes

$$g_c^\top D g_r = \frac{ad^2}{m}\left[w_1 am - w_2 b^2 + w_3 d(1-d)\right]. \tag{24}$$

The sign is therefore not invariant under arbitrary parameter scaling. At the original point, $D = \mathrm{diag}(2, 1, 1)$ changes the derivative from $-.0006$ to $+.0003$. More generally, a local parametrization $z = z(\theta)$ with Jacobian J induces $JJ^\top g_r$ under an ordinary θ-gradient step, giving success derivative $g_c^\top JJ^\top g_r$, not generally $g_c^\top g_r$. An orthonormal change of coordinates on the logit tangent space preserves the inner product; an arbitrary rescaling or neural-network parametrization does not.

At every interior point, g_r has signs $(+, -, -)$. Idealized first-step Adam applied to the full expected gradient with zero moments and negligible optimizer epsilon consequently gives success derivative $g_c^\top(1, -1, -1) = 2ad > 0$. At the original point this derivative is .06. This is not the expectation of stochastic Adam, since its nonlinear preconditioning does not commute with averaging. A fixed-score exponential tilt $p_\eta(i) \propto p_i \exp(\eta r_i)$ preserves the low-score conditional proportions and hence exact calibration. Its initial success derivative is

$$\mathrm{Cov}_p(c, r) = \mathrm{Var}_p(r) = a(1-a)(1-q)^2 = \frac{ad^2}{1-a} > 0. \tag{25}$$

At the original point the tilt derivative is .01. These alternatives delimit an ordinary logit-gradient claim; the family does not predict the actual optimizer trajectory of the source training system.

4 ProCredit: a score guarantee and its combined advantage

4.1 Published score and credit rules

ProCredit defines progress $\Phi_t \in [0,1]$ as the fraction of executable task checks passed after turn t (Ma et al., 2026, Section 3, PDF pp. 5–6). To avoid confusing correctness with the source's progress coefficient c, we write that coefficient as $b > 0$. With outcome $R \in \{0,1\}$,

$$\begin{aligned} r_t &= b(\Phi_t - \Phi_{t-1}), \\ S &= R + \sum_{t=1}^{T} r_t = R + b(\Phi_T - \Phi_0). \end{aligned} \tag{26}$$

For a common initial progress, bounded final progress, and $R = 1 \Rightarrow \Phi_T = 1$, a sure-success policy attains the maximum possible expected score. Under its availability, the score-optimal and success-optimal policy sets coincide (Ma et al., 2026, Appendix G, Proposition 3, PDF p. 23). This is a global statement about expected score, not every intermediate parameter update.

The source standardizes trajectory scores within a task group and adds centered turn-level returns:

$$\begin{aligned} A_i^{\text{traj}} &= \frac{S_i - \bar{S}}{s_G(S) + \epsilon_s}, \\ G_{i,t}^{(\gamma)} &= \gamma^{T_i - t} R_i + \sum_{u=t}^{T_i} \gamma^{u-t} r_{i,u}, \\ A_{i,t}^{\text{turn}} &= G_{i,t}^{(\gamma)} - \frac{1}{|\mathcal{U}|} \sum_{(j,u)\in\mathcal{U}} G_{j,u}^{(\gamma)}, \\ A_{i,t} &= A_i^{\text{traj}} + A_{i,t}^{\text{turn}}. \end{aligned} \tag{27}$$

Here $\mathcal{U}$ contains all turns of the sampled trajectories for the task. The source uses $\gamma = 1$, so $G_{i,t} = R_i + b(\Phi_{i,T_i} - \Phi_{i,t-1})$. Its clipped token-ratio objective averages over generated action tokens and excludes tool responses (source Eq. 2); no KL or entropy term is used. Reported defaults are $G = 8$ and $b = .5$ (Table I1, PDF p. 24).

4.2 One turn, one token, and exact pair enumeration

Consider ten initially unsatisfied task checks and one turn. Three one-token actions pass ten, one, or zero checks. Set

$$p = (.02, .75, .23), \quad c = (1, 0, 0), \quad \Phi_T = (1, .1, 0), \quad \Phi_0 = 0, \quad b = .5. \tag{28}$$

Then $S = (1.5, .05, 0)$. The full categorical simplex contains a sure-success vertex that uniquely maximizes score. Finite softmax logits parameterize its interior; we evaluate the local update at p.

With one turn per trajectory, $G_{i,1} = S_i$, and Eq. (27) reduces exactly to

$$A_i = \frac{S_i - \bar{S}}{s_G(S) + \epsilon_s} + (S_i - \bar{S}). \tag{29}$$

The source's token-mean objective is now the group mean. At the behavior policy, with inactive clipping and fixed sampled advantages, the gradient is $G^{-1} \sum_i A_i (e_i - p)$.

For $G = 2$, use population standard deviation, $\epsilon_s = 0$, and zero advantage on ties. In an ordered pair (i, j), define $d_{ij} = S_i - S_j$. Then

$$A_i = \text{sign}(d_{ij}) + \frac{d_{ij}}{2} = -A_j, \qquad g^{(i,j)} = \frac{A_i}{2}(e_i - e_j). \tag{30}$$

Summing the nine pairs with probabilities $p_i p_j$ gives

$$g_i = p_i \sum_j p_j \left[\text{sign}(S_i - S_j) + \frac{S_i - S_j}{2} \right]. \tag{31}$$

The resulting gradients are

$$\begin{aligned} g_c &= (.0196, -.015, -.0046), \\ g_S &= (.02865, -.013125, -.015525), \\ g &= (.033925, .1509375, -.1848625). \end{aligned} \tag{32}$$

Their directional products are

$$\begin{aligned} g_c^\top g_S &= .00082983 > 0, \\ g_c^\top g &= -.000748765 < 0, \\ g_S^\top g &= .001860886875 > 0. \end{aligned} \tag{33}$$

The raw-score gradient would improve success locally, but the expected combined-advantage direction reduces it while increasing mean score. The standardized trajectory term contributes $-.00116368$ to the success derivative; the centered turn term contributes $+.000414915$. The turn term mitigates the unfavorable standardized contribution. Both terms have the same sign within a pair; the reversal arises from their relative weighting across random groups.

Relation to existing normalization analyses. Bereket and Leskovec (2025) analyze overconfident probability prediction under proper scoring rewards; Wang (2026) studies amplification of policy-dependent prediction rewards in long-horizon agents. Both identify failures alleviated by removing standard-deviation normalization. The construction here fixes verified progress and examines ProCredit's specific advantage while preserving its global success-optimal policy set. It contrasts with Section 2, where mean centering leaves the unfavorable raw-gradient direction unchanged.

4.3 Default group size and removing standardization

For a group of size G, enumerate all count triples $n = (n_1, n_2, n_3)$ with $\sum_k n_k = G$. Their probabilities are

$$P(n) = \frac{G!}{n_1! n_2! n_3!} \prod_k p_k^{n_k}. \tag{34}$$

For each triple, compute the mean, standard deviation, and advantages in Eq. (29), then accumulate

$$\mathbb{E}[g] = \sum_n P(n) \sum_k \frac{n_k}{G} A_k(n)(e_k - p). \tag{35}$$

There are $\binom{G+2}{2}$ triples: 45 at the source's default $G = 8$. At $b = .5$ and $\epsilon_s = 10^{-6}$, the success derivatives are

Standard-deviation convention	$g_c^\top \mathbb{E}[g]$, $G = 8$
Population	$-.000929983748$
Sample	$-.000823033653$

The displayed source formula contains ϵ_s but does not fix its value or an unambiguous training standard-deviation convention. These are explicit sensitivity calculations, not a reconstruction of every implementation detail.

Removing the divisor from the trajectory term leaves two copies of $S_i - \bar{S}$. Exact mean-baseline enumeration gives

$$\mathbb{E}[g_{\text{no std}}] = 2\frac{G-1}{G} g_S. \tag{36}$$

The success derivative is positive for this construction: .00082983 at $G = 2$ and .0014522025 at $G = 8$. This intervention repairs the constructed ProCredit reversal, whereas Eq. (21) shows that it does not repair the calibrated SWE-RM family in Eq. (14). For that two-score family, both standardized and centered credit are positive multiples of g_r by Eqs. (19) and (21). Its mismatch is already present in the raw score; the ProCredit example instead acquires its unfavorable direction in the conversion to finite-group credit.

For fixed positive ϵ_s, the small score difference between the two failed actions vanishes as $b \to 0$, and the direction becomes favorable again. In the fixed-score large-G limit, Eq. (29) yields $[1/(\sigma_S+\epsilon_s)+1]g_S$, also favorable here. The example therefore does not assert reversal at arbitrary group sizes or coefficients. It neither refutes ProCredit's global score-optimum proposition nor reproduces AppWorld or ToolSandbox training. One turn also removes temporal-discount effects from this calculation.

5 Reproduction

Both programs use the Python standard library and run independently. Run `python reproduce_checks.py` in this folder to write `results.json`. It verifies:

- ARM-FM's two path returns and reversal thresholds;
- exact calibration, the conditional covariance, and the original ordinary-logit example;
- the GSPO limiting binomial expression against full count enumeration at $G = 2, 8, 16$;
- ProCredit's nine ordered pairs against an independent count representation, the 45 states at $G = 8$, and the no-standardization comparison.

Rational probabilities and nonstandardized expressions use Python's `Fraction`; standardized calculations use floating-point square roots. The output preserves the individual $G = 8$ count states and their probabilities, so the reported expectation can be inspected without Monte Carlo uncertainty.

Run `python` followed by `reproduce_calibration_family.py`; the output is `reproduce_calibration_family.results.json`. This program checks the family identities using exact rational arithmetic at 1,200 base policies, the strict negative-region formula at 6,000 numerical points, and fixed-score ambient directional derivatives by 80 central finite differences. It independently enumerates multinomial count vectors in 128 GSPO cases with $G \in \{2, 3, 8, 16\}$, population and sample standard deviations, and stabilizers 10^{-6} and .05, comparing the results with Eq. (18). It also checks the fixed-evaluator threshold, the slice root, a rectangle contained in the negative region, and the coordinate examples. These numerical points use the fixed seed

20261021. The analytical derivation establishes the identities over their stated domains; finite checks corroborate it and do not estimate prevalence in trained systems. This program does not rerun the ProCredit construction.

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